CSA5108 Cryptography and Network security

Exp 21 : ECB, CBC, and CFB modes
Code:

```
#include <stdio.h>

unsigned char E(unsigned char x, unsigned char k) {
   return x ^ k;
}

int main() {
   unsigned char p[] = {1, 2, 3, 4};
   unsigned char ecb[4], cbc[4], cfb[4];
   unsigned char key = 5, iv = 10, prev = iv;

   for (int i = 0; i < 4; i++) {
      ecb[i] = E(p[i], key);

      cbc[i] = E(p[i] ^ prev, key);
      prev = cbc[i];

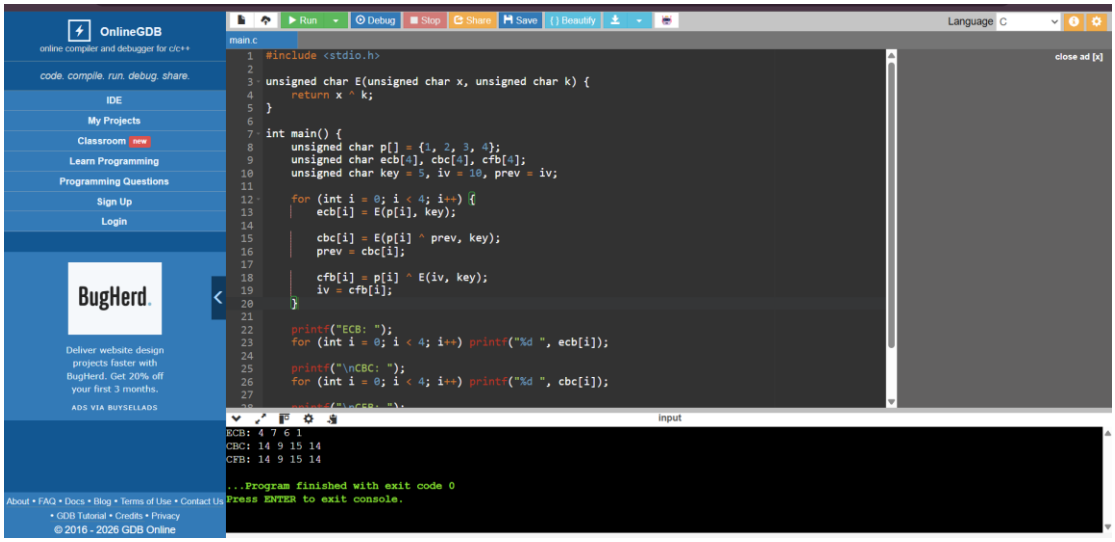
      cfb[i] = p[i] ^ E(iv, key);
      iv = cfb[i];
   }

   printf("ECB: ");
   for (int i = 0; i < 4; i++) printf("%d ", ecb[i]);

   printf("\nCBC: ");
   for (int i = 0; i < 4; i++) printf("%d ", cbc[i]);

   printf("\nCFB: ");
   for (int i = 0; i < 4; i++) printf("%d ", cfb[i]);
```

```
    return 0;
}
```


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Language C
main.c
close ad [x]
#include <stdio.h>
unsigned char E(unsigned char x, unsigned char k) {
return x ^ k;
}
int main() {
unsigned char p[] = {1, 2, 3, 4};
unsigned char ecb[4], cbc[4], cfb[4];
unsigned char key = 5, iv = 10, prev = iv;
for (int i = 0; i < 4; i++) {
ecb[i] = E(p[i], key);
cbc[i] = E(p[i] ^ prev, key);
prev = cbc[i];
cfb[i] = p[i] ^ E(iv, key);
iv = cfb[i];
}
printf("ECB: ");
for (int i = 0; i < 4; i++) printf("%d ", ecb[i]);
printf("\nCBC: ");
for (int i = 0; i < 4; i++) printf("%d ", cbc[i]);
input
ECB: 4 7 6 1
CBC: 14 9 15 14
CFB: 14 9 15 14
...Program finished with exit code 0
Press ENTER to exit console.